



AQ4.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Choose the set of correct options.

☐

If S is a spanning set of the vector space V , then $S \cup \{v\}$ must be a spanning set of V , for all $v \in V$.

☐

Span of an empty set is the zero vector space.

☐

If S is a spanning set of the vector space V , then $S \setminus \{v\}$ must be a spanning set of V , for all $v \in S$.

☐

If S is a spanning set of the vector space V , then $S \cup \{v\}$ may not be a spanning set of V , for all $v \in V$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If S is a spanning set of the vector space V , then $S \cup \{v\}$ must be a spanning set of V , for all $v \in V$.

Span of an empty set is the zero vector space.

Let S be the set of matrices where $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$. Consider the following statements:

- Statement P: $\text{Span}(S)$ is the vector space consisting of only lower triangular square matrices of order 2.
- Statement Q: $\text{Span}(S)$ is the vector space consisting of only upper triangular square matrices of order 2.
- Statement R: $\text{Span}(S)$ is the vector space consisting of all the square matrices of order 2.
- Statement S: $\text{Span}(S)$ is the vector space consisting of only scalar matrices of order 2.

Find the number of correct statements.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

If S is a basis of $\mathbb{R}^2$, then what is the cardinality of S ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point



Let $S = \{(1, 2), (1, 3)\}$ $S = \{(1, 2), (1, 3)\}$ and $\text{span}(S) = R^n$ $\text{span}(S) = R^n$. Then find the value of n .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Let S be the set of matrices where $S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ $S = \{[1000], [0001]\}$. Consider the following

statements:

- Statements P: $\text{Span}(S)$ is the vector space consisting of only the Identity matrix of order 2.
- Statements Q: $\text{Span}(S)$ is the vector space consisting of only diagonal matrices of order 2.
- Statements R: $\text{Span}(S)$ is the vector space consisting of all the square matrices of order 2.
- Statements S: $\text{Span}(S)$ is the vector space consisting of only scalar matrices of order 2.

Find the number of correct statements.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

1 point

Consider the following set of vectors $S = \{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$ $S = \{(1, 1, -1), (-1, 1, 1), (0, 2, 1, 0), (0, 1, -2), (1, 0, -2)\}$. Choose the set of correct options.

☐

The set $\{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0)\}$ is not a basis of R^3 .

☐

The set $\{(0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$ is a basis of R^3 .

☐

The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a basis of R^3 .

☐

None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0)\}$ is not a basis of R^3 .

The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a basis of R^3 .

1 point

Consider a set of vectors $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\}$ $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\}$ in R^2 . Choose the set of correct options.

☐

The set $\{(1, 2), (-1, 2), (3, 1)\}$ is a basis of R^2 .

☐

The set $\{(-1, 2), (3, 1)\}$ is a basis of R^2 .

☐

The set $\{(3, 1), (-3, 1)\}$ is a basis of R^2 .

☐

None of the above.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(-1, 2), (3, 1)\}$ is a basis of R^2 .

The set $\{(3, 1), (-3, 1)\}$ is a basis of R^2 .

Level 2

1 point

Choose the set of correct options.

☐

If a subset SS of R^2 contains only two elements then SS is a basis of R^2 .

☐

If a subset SS of R^3 contains only one elements then SS can never be a basis of R^3 .

☐

If a subset SS of RR contains only one non zero element then SS is a basis of RR .

☐

If S_1 and S_2 are two bases of R^3 then $S_1 = S_2$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If a subset SS of R^3 contains only one elements then SS can never be a basis of R^3 .

If a subset SS of RR contains only one non zero element then SS is a basis of RR .

1 point

Let V be the subspace of R^3 defined as follows:

$$V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in R\}$$

Choose the set of correct options from the following.

☐

$\{(1, 1, 0), (1, 0, -1)\}$ is a linearly independent set of V .

☐

$\{(1, 1, 0), (1, 0, -1), (0, 1, 1)\}$ is a linearly independent set of V .

☐

$\{(0, 1, 1), (1, 0, -1)\}$ is a spanning set of V .

☐

$\{(1, 1, 0)\}$ is a spanning set of V .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(1, 1, 0), (1, 0, -1)\}$ is a linearly independent set of V .

$\{(0, 1, 1), (1, 0, -1)\}$ is a spanning set of V .

1 point

Let V and W be vector spaces which are defined as follows:

$$V = \{(x, y) \mid y = mx, \text{ where } m \neq 0 \text{ and } x, y, m \in R\}$$

$\{(x, y) \mid y = mx, \text{ where } m = 0 \text{ and } x, y, m \in R\}$ with usual addition and scalar multiplication as in

R^2 . $W = \{(x, y) \mid x = 0\}$ with usual addition and scalar multiplication as in R^2 .

$\{(x, y) \mid x = 0\}$ with usual addition and scalar multiplication as in R^2 .

Choose the correct set of options.

☐

The set $\{(1, m), (\frac{1}{m}, 1)\}$ is a linearly independent set in V .

☐

The set $\{(1, m), (\frac{1}{m}, 1)\}$ is a spanning set for V .

☐

The set $\{(1, m)\}$ is a linearly independent set in V .

☐

The set $\{(\frac{1}{m}, 1)\}$ is a linearly independent set in V .

☐

The set $\{(0, 1), (0, 2)\}$ is a linearly independent set in W .

☐

The set $\{(0, 1)\}$ is a linearly independent set in W .

☐

The set $\{(0, 5)\}$ is a spanning set for W .

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{(1, m), (\frac{1}{m}, 1)\}$ is a spanning set for V .

The set $\{(1, m)\}$ is a linearly independent set in V .

The set $\{(\frac{1}{m}, 1)\}$ is a linearly independent set in V .

The set $\{(0, 1)\}$ is a linearly independent set in W .

The set $\{(0, 5)\}$ is a spanning set for W .

[Check Answers](#)

Your score is: 0/10